

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 15E.8

SPLITTING PATTERNS IN ^1H NMR SPECTRA

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: CARBOXYLIC ACID DERIVATIVES, SPECTROSCOPY AND CHROMATOGRAPHY
SUB-TOPIC:	15E.8: Splitting Patterns in ^1H NMR Spectra
TOTAL QUESTIONS:	2 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	5 Marks
EXAMINER FAQs:	1 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 15E.8

Q1. 1. (a) Apply the n+1 rule to explain the splitting pattern observed in the ^1H NMR spectrum of ethyl ethanoate, $\text{CH}_3\text{COOCH}_2\text{CH}_3$.

[4 Marks]

WCH14/01/Jun23/Q10(a)

Q2. (b) State why $-\text{OH}$ and $-\text{NH}$ protons usually appear as singlets regardless of adjacent protons.

[1 Marks]

WCH14/01/Jun23/Q10(b)

SECTION B: WORKED MARK SCHEME — 15E.8

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. (a) Apply the n+1 rule to explain the splitting pattern observed in the ^1H NMR spectrum of ethyl ethanoate, $\text{CH}_3\text{COOCH}_2\text{CH}_3$. 1. CH_3 of ethanoate ($\text{CH}_3\text{CO}-$): singlet (0 adjacent H atoms, $n=0 \rightarrow 1$ peak). 2. $-\text{CH}_2-$ of ethyl ($-\text{OCH}_2\text{CH}_3$): quartet (split by 3 adjacent H of CH_3 , $n=3 \rightarrow 4$ peaks, 1:3:3:1 ratio). 3. $-\text{CH}_3$ of ethyl ($-\text{OCH}_2\text{CH}_3$): triplet (split by 2 adjacent H of CH_2 , $n=2 \rightarrow 3$ peaks, 1:2:1 ratio).	[4]
Q2	Q2: (b) State why $-\text{OH}$ and $-\text{NH}$ protons usually appear as singlets regardless of adjacent protons. Rapid proton exchange with trace water/solvents averages out spin-spin coupling.	[1]

SECTION C: EMPIRICAL EXAMINER FAQs & COMMON TRAPS — 15E.8**FAQ #1: What is the n+1 rule in ^1H NMR?****Category: Spin-Spin Coupling****COMMON EXAMINER TRAP / MISTAKE:**

Counting protons on the SAME carbon atom.

MODEL ANSWER / EXAMINER EXPECTATION:

The number of peaks in a multiplet is $n+1$, where n is the number of hydrogen atoms on ADJACENT carbon atoms (none-equivalent protons 3 bonds away).